

Fig. 1: WSN building blocks

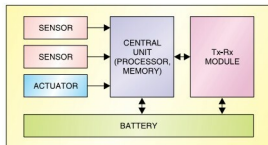


Fig. 2: Structure of a sensor node

the placement of devices, the configuration of parameters, application binding, optimisation of network and device parameters, and testing and verification of correct operation. Easily configurable and maintainable systems, ease of identification and accessibility of devices, and interoperability and co-existence with other wired and wireless systems are some of the major focus areas while commissioning any wireless sensor network.

Interference. The best way to get rid of the effects of RF interference is to place the wireless sensor network on a different channel. This can be as simple as manually configuring the network to the alternate channel or enabling the network to perform an intelligent channel scan/start-up to look for the clearest channel. The effects of difficult building environments can be mitigated either by the use of a

high-power, high-sensitivity module to gain a significant improvement in link budget or with additional router nodes in a mesh network, placed to bypass particular obstacles. Packet error rate testing and link-quality measurements can be effective in understanding the environment, identifying the clearest

RF channel and determining the best packet size for transmission (in order to minimise packet error rates).

Embedded platform selection

A WSN is a large-scale distributed architecture that incorporates hardware and software elements. Therefore both hardware and software must be simultaneously considered during the design phase of any WSN. The factors to consider during the selection process include transmit power, receiver sensitivity, antenna types, physical layer, data link layer, network layer and transport layer.

As shown in Fig. 1, generally, wireless sensor networks use three types of devices: coordinator, repeaters/routers and sensor nodes. All these basic networking elements will usually be driven by 8-bit microcontrollers, but

depending on the complexity and the network they are connected to, they would require 16-bit or low-end 32-bit microcontrollers.

The hardware platform. Diverse non-uniform embedded hardware devices with both real-time performance and low power are available for use in wireless sensor networks. This section discusses the basic features of WSN components and the factors to consider while choosing them.

Wireless module. Each of the coordinator, end device and repeater/router units is equipped with a wireless module or RF engine, to receive and transmit packets of data. It also checks and recovers errors. In the case of secure applications, it will decrypt the received data and encrypt any data to be transmitted.

Many semiconductor vendors ship IEEE 802.15.4-compatible radios. Some of them also sell SoCs, SiPs or single-chip solutions that integrate an IEEE 802.15.4 radio to realise the PHY and MAC layers, MCU core and standard input/output (I/O) peripheral units. Typically, products of this type are available from Ember, Jennie, Zensys and Texas Instruments.

Sensor nodes. These are the main workhorses. Typically, a WSN consists of a large number of small sensor nodes spread out in areas that are difficult to access. These nodes communicate wirelessly. They should combine the abilities to compute, communicate, acquire real-world information and drive actuators in a single-chip solution, as shown in Fig. 2.

Low-cost CPUs that consume less power and have small memory footprints often characterise the sensor nodes used for relatively simpler functions. The limited resources of the hardware create special requirements for the system software and the applications that need to run on it. Many offerings are based on well-known processors like 8051, Arm PIC or Rabbit. These use well-established integrated development environments, compilers and debuggers.

The transmitter-receiver (Tx-Rx)